# Chapter 4

# Results and discussion

Draft for research-team review · version 2026-10-20-v01 · revision timestamp 20 October 2026, 16:13 PHT (UTC+08:00) · incorporates the dose-response, scaled-error contribution, exact-completeness and full-read editorial updates; existing experimental results only

## 4.1 Scope and principal finding

This chapter reports the completed synthetic-data experiments for the AICS case reference prototype. The fixed main test **failed**. On the 36 non-transport reviewed cases at p = .50 and five retrieved neighbors, equal-weight retrieval had mean absolute error (MAE) of **₱1,903.82**, compared with **₱1,333.76** for the same-purpose baseline. The resulting reduction was **-42.74%**, rather than the required reduction of at least 20%. The maximum MAE compatible with that criterion was ₱1,067.01. These results do not establish that the proposed retrieval method works on this dataset under the study's stated success criterion.

Separately, all four non-medical assistance types show steadily increasing reductions in case-base retrieval error relative to their condition-specific purpose baselines as planted household signal increases; this ordered pattern is descriptive and does not change the fixed reviewed-case failure.

The separate weight-learning experiment improved over equal weights under its specified tests. That finding addresses a different comparison and does not replace the failed main test. In particular, improvement also appeared in the no-household-signal control. It cannot, by itself, establish recovery of household structure or usefulness in real benefit decisions.

All records and grants in these experiments are synthetic. The forty reviewed intake sheets are fixed test cases; their evaluated grants were generated by the written rule, not supplied here as new social-worker judgments. This draft stops at Chapter 4. A separate manuscript copy fills only the six environment blanks in Section 3.2.4.3; the approved retrieval specification is unchanged. No team approval of additional implementation choices is claimed.

## 4.2 Implementation and provenance

The comparison uses the 39 fields specified in the manuscript, same-type reference pools, applicant-level exclusion, reference-only numeric ranges and constants, Jaccard scores for multi-select fields, and the specified treatment of unknowns, not-applicable values and shared absences. All events of a held-out applicant are excluded. The reviewed cases are never inserted into the case base; the evaluator also applies their documented partner exclusions. The prototype displays five past cases using equal weights and does not output a prediction or applicant score.

The reconciled implementation restores the supplied amount-code scoring semantics and sorted-payload noise algorithm. Compatibility tests reproduce all 160 supplied forty-case amounts with the supplied calibration. The study run instead uses the required calibration from the 1,000-event case base and reuses it for the forty test cases. Per-event scoring avoids the source script's repeated-applicant overwrite. Cost and rounding constraints are enforced explicitly.

Researcher-selected cost ranges were retained from the earlier preparation, not changed in response to its failed result. They use uniform intervals from 0.5 to 1.5 times available cost benchmarks; device and transport ranges remain declared assumptions. Educational costs use a tuition proxy. Laboratory uses the published ratio .9052; dialysis and chemotherapy each use .3556 from their combined source row. These completions and proxies must remain disclosed in the methodology.

The existing base had 244 wage-label contradictions. The reconciled copy removes the contradictory labels without changing incomes or worker counts. New generated sets enforce that consistency before inserting unknown values. Ten original-base events remain unverifiable because worker counts are missing; they are flagged, not presented as validated. Unknown values are still excluded from Gower. The source amount rule's distinct missing-score semantics are preserved and are a limitation of the synthetic mechanism.

The earlier provisional run is retained separately: its main-test retrieval MAE was ₱1,930.18 against ₱923.49, a −109.01% reduction. The reconciled run is a distinct version justified by source compatibility, not a replacement advertised as success.

The purpose-baseline MAE also rose from ₱923.49 in the provisional run to ₱1,333.76 in the reconciled run (44.43%); reconciliation changed the generated outcomes and calibration used by the comparator, so this is a change of run definition, not an improvement measured against a fixed baseline.

## 4.3 Data composition and published benchmarks

The primary base contains 1,000 events from 844 applicant IDs. The published-mix column below is the comparator recorded in the supplied figures module (Table 12); the synthetic mix was deliberately flattened. It is not a sampled Davao City population distribution. Rounded published percentages sum to approximately 100%.

| Assistance type | Generated events | Generated % | Bundle published comparator % |
|---|---|---|---|
| Cash Relief Assistance | 100 | 10.0 | 8.3 |
| Educational | 80 | 8.0 | 4.0 |
| Food Assistance | 80 | 8.0 | 3.8 |
| Funeral | 150 | 15.0 | 12.3 |
| Medical | 510 | 51.0 | 70.6 |
| Transportation | 80 | 8.0 | 1.1 |

The flattened mix cuts against the study’s interpretation: medical represents 51.0% of generated events versus 70.6% in the supplied published comparator. Because medical has the largest error and baseline deficit, a more medical-heavy mix would worsen the pooled result if the observed within-type errors held fixed. This is a composition sensitivity interpretation, not an estimate from a representative population or a newly generated sample.

Six-month income bands are retained as generated, without balancing them to improve detectability.

| Income band | Base events | Reviewed cases |
|---|---|---|
| Below food threshold | 529 | 26 |
| Food to poverty threshold | 148 | 7 |
| At or above poverty threshold | 271 | 7 |
| Unknown | 52 | 0 |

The cost checks below compare generated means with the historical published anchors, not current benefit policy. No post-result cost adjustment was made.

| Group | Published mean cost ₱ | Generated mean cost ₱ | Difference % | Within 20% |
|---|---|---|---|---|
| Hospital bill | 147,375.00 | 150,927.42 | 2.41 | True |
| Funeral | 45,230.74 | 46,048.33 | 1.81 | True |
| Medicines | 7,831.71 | 7,949.84 | 1.51 | True |
| Laboratory | 5,800.00 | 6,040.52 | 4.15 | True |
| Educational | 11,700.00 | 11,261.25 | -3.75 | True |
| Chemotherapy | 23,142.86 | 21,652.78 | -6.44 | True |
| Dialysis | 23,142.86 | 24,555.26 | 6.10 | True |

The following p = .50 summaries compare aggregate generated shares (total grants divided by total costs) with the declared share bases. Food and cash relief instead compare mean amounts in pesos. Shares are not mean individual ratios. The borrowed device share and exact transport share are assumptions, not Region XI estimates.

| Purpose | n | Generated share or mean ₱ | Rule base share or mean ₱ |
|---|---|---|---|
| Allowance | 19 | 0.6540 | 0.6105 |
| Assistive device | 14 | 0.4009 | 0.3556 |
| Cadaver transfer | 16 | 0.3313 | 0.2903 |
| Chemotherapy | 18 | 0.3525 | 0.3556 |
| Dialysis | 19 | 0.3479 | 0.3556 |
| Food subsidy | 80 | 4,096.25 | 3,857.14 |
| Funeral expenses | 134 | 0.3057 | 0.2903 |
| Hospital bill | 93 | 0.2121 | 0.2075 |
| Laboratory | 58 | 0.8163 | 0.9052 |
| Medicines | 308 | 0.7780 | 0.8046 |
| Other needs | 100 | 4,687.50 | 4,400.00 |
| School fees | 35 | 0.6381 | 0.6105 |
| School supplies | 26 | 0.6223 | 0.6105 |
| Transportation for medical treatment | 33 | 1.0000 | 1.0000 |
| Transportation to home province | 47 | 1.0000 | 1.0000 |

The published amounts and costs were reported for a historical population and may include payments by other sources. Here documented cost is the outstanding need. Agreement of selected summary means therefore supports a limited plausibility check, not population validity or proof of realistic joint relationships.

## 4.4 Missingness and effective comparisons

Unknown entries are not imputed into the distance. Explicit No and not-applicable values follow the manuscript's separate rules. The primary base has the following unknown counts among matching fields; rates use 1,000 events as the denominator.

| Matching field | Unknown events | Unknown % |
|---|---|---|
| client_type | 0 | 0.0 |
| crisis_catastrophic_event | 21 | 2.1 |
| crisis_death_of_member | 14 | 1.4 |
| crisis_disablement | 19 | 1.9 |
| crisis_hospitalization | 28 | 2.8 |
| crisis_last_3_months | 27 | 2.7 |
| crisis_loss_of_livelihood | 25 | 2.5 |
| crisis_others | 15 | 1.5 |
| disability_type | 2 | 0.2 |
| emergency_fund | 23 | 2.3 |
| employed_members | 14 | 1.4 |
| ext_guarantee_letter_other_agency | 19 | 1.9 |
| ext_health_card | 18 | 1.8 |
| ext_mss_discount | 23 | 2.3 |
| ext_philhealth | 26 | 2.6 |
| ext_pwd_discount | 24 | 2.4 |
| ext_senior_citizen_discount | 17 | 1.7 |
| illness_duration | 14 | 1.4 |
| insurance_coverage | 21 | 2.1 |
| monthly_expenses | 45 | 4.5 |
| occupation | 15 | 1.5 |
| purpose_of_assistance | 0 | 0.0 |
| referral_type | 23 | 2.3 |
| savings | 19 | 1.9 |
| seasonal_members | 19 | 1.9 |
| selfhelp_reached_agencies | 24 | 2.4 |
| selfhelp_sought_income | 22 | 2.2 |
| service_setting | 21 | 2.1 |
| sub_category | 22 | 2.2 |
| support_church | 18 | 1.8 |
| support_employer | 28 | 2.8 |
| support_family | 17 | 1.7 |
| support_friends | 12 | 1.2 |
| support_relatives | 15 | 1.5 |
| target_sector | 0 | 0.0 |
| total_income_past_6_months | 52 | 5.2 |
| vuln_elderly_child_pwd_pregnant | 18 | 1.8 |
| vuln_member_incapacitated | 20 | 2.0 |
| vuln_unstable_employment | 32 | 3.2 |

At p = .50, the equal-weight retrieved pairs compared between 24 and 36 fields (median 31). Learned-weight retrieved pairs had between 21 and 34 comparable fields with positive weight (median 28). The equal-weight primary run counted 0 undefined eligible pairs. The machine-readable results retain each neighbor's count; available-field counts and positive-weight counts are distinguished.

The source schema lacks the separate Section V not-applicable tick, so all-No resource rows cannot independently establish whether that tick was recorded. The supplied explicit No values are retained with a provenance warning. This cannot be repaired by inferring an unrecorded answer.

## Exact record-level completeness

Counting the existing CSV records directly gives the following completeness figures. Unknown means an empty or explicitly unrecorded value under the existing loader; explicit No, numeric zero and recorded not-applicable values are not counted as unknown. These are event counts, not counts of distinct applicants.

| Dataset | Field scope | Events | Unknown cells | Complete events | Incomplete events |
|---|---|---|---|---|---|
| Primary case base | 39 matching fields | 1000 | 772 | 463 | 537 |
| Primary case base | 28 household-rule fields | 1000 | 606 | 545 | 455 |
| Frozen reviewed cases | 39 matching fields | 40 | 21 | 22 | 18 |
| Frozen reviewed cases | 28 household-rule fields | 40 | 14 | 27 | 13 |

For the 39 matching fields, 463 of 1,000 events (46.3%) are complete. The 772 unknown cells give a mean cell-level rate of 1.98%; multiplying the per-field observed proportions under an independence assumption would predict approximately 457.6 complete events, but the actual joint count is 463. For the correction to the household score, the relevant direct count is narrower: 455 events (45.5%) have an unknown in at least one of its 28 fields, containing 606 unknown cells. The other 545 events have complete household-rule inputs. Missingness is therefore material, even though the typical per-field rate is small.

Across the frozen 40-case source file, excluding the blank outcome column, there are 287 unknown cells in 21 columns and every case has at least one. That broader scope includes context and income-source fields not used in matching or the household score. Within the 39 matching fields, only 18 cases have unknowns (21 cells); within the 28 household-rule fields, 13 have unknowns (14 cells). The statement that all 40 source cases contain an unrecorded entry must not be read as saying all 40 household scores are incomplete.

The reviewed inputs remain frozen, and removing unknowns from the case-base generator would alter the data design rather than address the observed missingness. Any approved correction must handle existing unknowns explicitly. The proposed shifted score preserves a complete record's raw score, but recalibration can change standardized scores and generated amounts for complete records as well. It does not follow that every amount must change, because rounding and bounds may preserve some. A corrected-data run would be a new, prospectively approved version with these reasons disclosed, while every prior run remains retained. No such run is included here.

## 4.5 Separate-set selection of money and duration treatments

A separate primary set used master seed 41000; thirty new sets used 41001–41030. All selection comparisons used equal weights and p = .50. Money transformations were compared first, then duration under the selected money treatment. The criterion used applicant-averaged scaled absolute errors, with transportation excluded because it has no scaled width and its grant is exactly its ticket cost. The noise floor was the sample SD of the thirty paired mean gaps. Each challenger also had to pass the primary-set two-sided Wilcoxon test, with α = .025 for money and α = .05 for duration. Applicants, not a pooled stack of replicated IDs, were the paired units. All-zero differences would give p = 1.

| Comparison | Mean scaled improvement | Noise-floor SD | Wilcoxon p | Passes both |
|---|---|---|---|---|
| Log versus plain | 0.001083 | 0.002761 | 0.268 | False |
| Fuzzy versus plain | 0.002880 | 0.002999 | 0.849 | False |
| Rank versus categorical duration | 0.000394 | 0.001464 | 0.195 | False |

The retained settings are **plain money scaling** and **categorical duration**. Failure to pass is not proof of equivalent methods; it means the preregistered implementation criterion did not justify replacing the baseline.

For transparency, these are medians of the query-specific reference-pool income quartiles on the separate primary set. Each actual query used its own eligible pool. The signed percentage differences compare Q2 with the food edge and Q3 with the poverty edge.

| Type | Q1 ₱ | Q2 ₱ | Q3 ₱ | Q2–food edge % | Q3–poverty edge % |
|---|---|---|---|---|---|
| Cash Relief Assistance | 24,900.00 | 46,950.00 | 83,300.00 | -14.89 | 5.71 |
| Educational | 16,300.00 | 43,900.00 | 77,700.00 | -20.42 | -1.40 |
| Food Assistance | 17,325.00 | 44,600.00 | 80,200.00 | -19.15 | 1.77 |
| Funeral | 12,850.00 | 40,750.00 | 71,400.00 | -26.13 | -9.39 |
| Medical | 18,525.00 | 43,300.00 | 86,625.00 | -21.50 | 9.93 |
| Transportation | 13,150.00 | 33,200.00 | 81,700.00 | -39.81 | 3.68 |

Fuzzy membership used linear low/middle/high memberships anchored at these quartiles, with similarity equal to one minus half their L1 distance. Coincident quartiles triggered a recorded plain-range fallback. These are explicitly documented implementation choices, not formulas supplied verbatim in the paper.

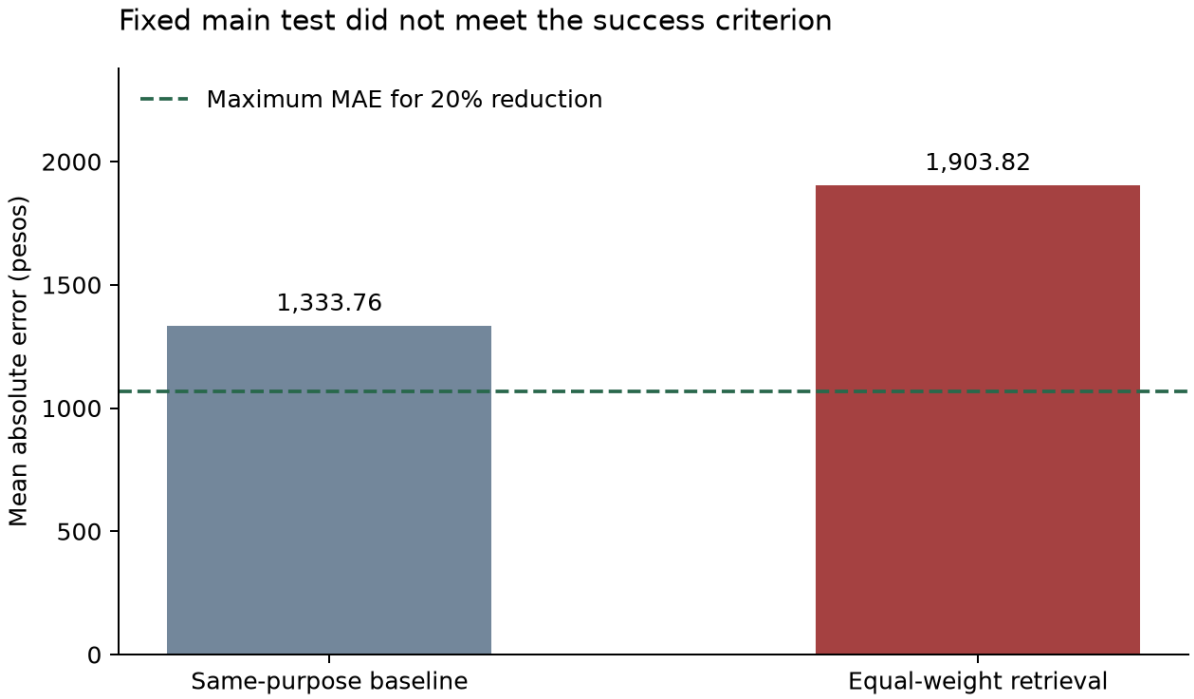
## 4.6 Fixed reviewed-case test

For cost-bearing cases, retrieval combines the five recorded grant-to-cost ratios by their median and applies it to the query's own cost. Direct food/cash amounts are combined by their median. The baseline uses the same-purpose eligible pool with the same exclusions. The following p = .50 results are descriptive by type; the success criterion is the pooled 36-case peso comparison, not a type-specific or scaled-error substitute.

| Type | Cases | Retrieval MAE ₱ | Purpose-baseline MAE ₱ | Reduction % | Scaled MAE |
|---|---|---|---|---|---|
| Cash Relief Assistance | 5 | 590.00 | 520.00 | -13.46 | 0.065556 |
| Educational | 5 | 1,116.86 | 1,068.97 | -4.48 | 0.248191 |
| Food Assistance | 4 | 325.00 | 512.50 | 36.59 | 0.162500 |
| Funeral | 8 | 1,648.39 | 2,612.43 | 36.90 | 0.082419 |
| Medical | 14 | 3,251.16 | 1,222.94 | -165.85 | 0.055393 |
| Transportation | 4 | 0.00 | 0.00 | N/A | N/A |

The four transportation cases have zero error by construction. They are reported separately and do not count toward the main test. Plain and selected settings coincide in this run. Results across all conditions are:

| Household condition | 36-case MAE ₱ | Purpose-baseline MAE ₱ | Reduction % |
|---|---|---|---|
| p00 | 2,205.44 | 1,805.07 | -22.18 |
| p25 | 2,096.81 | 1,361.40 | -54.02 |
| p50 | 1,903.82 | 1,333.76 | -42.74 |
| p75 | 1,542.06 | 1,127.28 | -36.79 |




## Medical error diagnosis

Medical assistance accounts for 14 of 36 cases (38.9%) but **66.4% of total retrieval absolute error**. Its retrieval MAE is ₱3,251.16 against ₱1,222.94 for the purpose baseline, a −165.85% reduction. Medical is the source of the aggregate failure: the remaining 22 cases have retrieval MAE **₱1,046.43** against **₱1,404.29**, a **25.48% reduction**.

| Diagnostic subset | Cases | Retrieval MAE ₱ | Purpose-baseline MAE ₱ | Reduction % |
|---|---|---|---|---|
| All non-transport cases — fixed main test | 36 | 1,903.82 | 1,333.76 | −42.74 |
| Medical | 14 | 3,251.16 | 1,222.94 | −165.85 |
| Non-medical non-transport — post-result diagnosis | 22 | 1,046.43 | 1,404.29 | 25.48 |

The 22-case subset exceeds 20% descriptively; it is not a replacement success test and does not change the failed 36-case conclusion. Cash relief and education individually remain below their baselines. The diagnosis concerns the aggregate concentration of error, not a claim that every non-medical type passes.

| Assistance type | Purpose and base share or direct amount |
|---|---|
| Medical | Hospital bill .2075; dialysis .3556; chemotherapy .3556; assistive device .3556; medicines .8046; laboratory .9052 |
| Funeral | Funeral expenses and cadaver transfer: .2903 each |
| Educational | Allowance, school fees and school supplies: .6105 each |
| Food assistance | Single purpose, food subsidy: direct base ₱3,857.14 |
| Cash relief | Single purpose, other needs: direct base ₱4,400.00 |
| Transportation | Both purposes: 1.0000; exact cost with no household adjustment |

Medical is the only type with different purpose-specific base shares, spanning .2075–.9052. The baseline is conditioned on purpose, while retrieval pools by type and gives purpose one of 39 equal nominal weights (1/39, about 2.6%; the effective share depends on the available-field denominator). A medicines query can therefore retrieve hospital-bill cases with a much lower base share. This is the mechanism supported by the error concentration and the known rule; the existing comparisons do not isolate its causal contribution from every other generator limitation.

The separate case-base experiment corroborates the role of purpose: its mean learned weight is 2.9797, compared with 1.5750 for the next field. At p = .50, medical's applicant-averaged reduction changes from −340.42% under equal weights to −20.36% under learned weights, the largest type-specific improvement. These are case-base results, not the 14 reviewed-case results above.

## Retrieval error across household-signal levels

Appendix 4A also supports an ordered comparison across the four planted household-signal conditions. The following figures are case-base results: errors are first averaged within applicant and assistance type across the 1,000 events, then across applicants in that type. They are distinct from the fixed 36 reviewed-case test. Each percentage uses its own condition-specific purpose baseline, which changes because the generated amounts change with the condition. Transportation is excluded here because its error is zero by construction.

### Equal weights — reduction against the matching purpose baseline (%)

| Assistance type | p00 | p25 | p50 | p75 | Strictly increasing |
|---|---|---|---|---|---|
| Cash relief | -14.28 | -10.41 | +0.66 | +10.02 | Yes |
| Educational | -3.33 | -1.83 | +9.89 | +16.59 | Yes |
| Food | -13.50 | -12.94 | -1.35 | +23.02 | Yes |
| Funeral | -7.31 | -5.15 | +3.47 | +14.81 | Yes |
| Medical | -360.43 | -387.41 | -340.42 | -332.69 | No |

For each of the four non-medical types, the equal-weight reduction increases at every step as the household share of variation rises from 0% to 75%. All four trail their purpose baseline at p00, where there is no household signal, and all four exceed it at p75. This dose-response pattern is consistent with retrieval responding to planted household information where the medical purpose-share mismatch is absent. Medical remains far below its baseline throughout; its equal-weight sequence is non-monotonic, rather than literally constant or evidence of statistical equivalence.

### Learned weights — reduction against the matching purpose baseline (%)

| Assistance type | p00 | p25 | p50 | p75 | Strictly increasing |
|---|---|---|---|---|---|
| Cash relief | -12.28 | -15.09 | +2.31 | +13.35 | No |
| Educational | -4.51 | -16.58 | +5.04 | +12.30 | No |
| Food | -24.08 | -3.75 | +2.54 | +16.53 | Yes |
| Funeral | -11.81 | -4.43 | +13.06 | +14.44 | Yes |
| Medical | -35.11 | -27.08 | -20.36 | +2.21 | Yes |

Under learned weights, medical improves monotonically from −35.11% at p00 through −27.08% and −20.36% to +2.21% at p75. This is consistent with emphasizing purpose reducing an obstacle to observing household-signal response. It does not isolate the effect of purpose from all other learned-weight changes. Food and funeral also increase monotonically. Cash relief and education dip at p25, so the learned-weight results must not be described as monotonic for every type.

These findings are compatible with Section 4.8. The weight search minimizes an applicant-averaged scaled-error objective across the whole case base and assigns its largest mean weight to purpose. Medical dominates the unscaled peso error under equal weights, but that alone does not establish its influence on the scaled optimization objective. The scaled-error breakdown below addresses that separate question using the saved errors. The observed weights and error changes support purpose as a major lever used by the search. They do not demonstrate additional household-pattern recovery by the search. Retrieval can nevertheless respond to household signal through fields already included under equal weights, as the non-medical ordered comparisons show. What the search emphasizes and what retrieval responds to are different findings.

There are only four points per type. This is a descriptive pattern, not a formal trend test; no statistical significance, fitted trend line or causal proof is claimed. These comparisons do not isolate the mechanism from the generator's other limitations. The main condition remains p = .50, fixed before evaluation. The better p75 results do not justify choosing a different main condition. The fixed reviewed-case test still failed at −42.74%, as Section 4.1 reports; this case-base analysis explains aspects of that result and does not overturn or replace it. It supports a household-signal response in the four non-medical types and a purpose-confusion explanation for medical that remains to be isolated in an approved comparison.

## Scaled-error contribution and the observed trade-off

The scaled results allow a more direct check of medical's contribution. Multiplying each type's mean scaled error by its applicant count gives the approximate proxy below. Applicants appearing in more than one type are counted in more than one row, so that proxy is not a partition of the actual applicant-averaged objective.

| Type | Applicants in type | Equal scaled error | Overlapping-row proxy share % | Exact held-out objective share % |
|---|---|---|---|---|
| Medical | 461 | 0.130962 | 41.4 | 43.34 |
| Food Assistance | 80 | 0.423750 | 23.2 | 22.14 |
| Educational | 78 | 0.307455 | 16.4 | 16.25 |
| Funeral | 143 | 0.122569 | 12.0 | 11.61 |
| Cash Relief Assistance | 98 | 0.102381 | 6.9 | 6.67 |

The exact column is reconstructed from the saved p = .50 held-out event errors, with no retrieval or fitting rerun. For each applicant a, divide each non-transport event error by that applicant's count of non-transport events; then sum those fractions by type and divide by the 794 applicants with a defined scaled error (844 total applicant IDs minus 50 transportation-only applicants, whose events have no scaled width). The contributions sum to the recorded equal-weight objective of 0.167927. Medical contributes 0.072774, or **43.34%**, the largest single block; the overlapping-row approximation is **41.4%**. These are the held-out errors under the fitted folds, not the training objective values evaluated inside each search. The same applicant-averaging convention underlies both, but the eligible reference pools differ.

| Type | Equal scaled error | Learned scaled error | Learned minus equal |
|---|---|---|---|
| Medical | 0.130962 | 0.080893 | -0.050069 |
| Food Assistance | 0.423750 | 0.407500 | -0.016250 |
| Educational | 0.307455 | 0.324025 | +0.016570 |
| Funeral | 0.122569 | 0.110388 | -0.012181 |
| Cash Relief Assistance | 0.102381 | 0.100680 | -0.001701 |

Medical has the largest decrease in per-type applicant-averaged scaled error, about 0.0501, roughly 3.1 times the next largest decrease (food, about 0.0163). Educational error instead rises by about 0.0166. The exact objective decomposition likewise assigns the largest reduction to medical: its contribution falls by 0.027421, while education's rises by 0.000915. The shared weights therefore produce an observed trade-off: substantially lower medical error accompanies higher educational error.

This is consistent with purpose being a lever used by the search and medical being where that lever paid off. It describes the outcome of the learned weights; it does not isolate the optimizer's internal reasons, prove that education was deliberately sacrificed, or establish that purpose was its only useful lever. Section 4.8's household-recovery conclusion remains unchanged.

## 4.7 Five-fold learned weights and thirty-set evidence

Applicant IDs were assigned to five fixed folds. Each fold's shared 39 weights were fitted only on the other four folds, using applicant leave-one-out retrieval inside that training pool and 200 objective evaluations. All weights were nonnegative and summed to 39. Held-out applicants retrieved from all other case-base applicants, including other applicants assigned to their test fold; the equal-weight comparator used that identical reference pool. Settings selected on the separate set were held fixed. No reviewed case selected weights or budget.

Thirty weight replicates used master seeds 51001–51030. The same two-part evidence criterion was applied with $\alpha = .05$ per specified condition. The paired test used the primary case-base applicants. The following peso MAEs first average the non-transport event errors within each applicant, then average applicants; scaled errors similarly average each event's error divided by its prescribed band width. This differs from an event-weighted pooled MAE when applicants have repeat visits.

| Condition | Equal MAE ₱ | Learned MAE ₱ | Paired mean peso gain | Purpose baseline ₱ | 30-set scaled gain | SD | Wilcoxon p | Passes |
|---|---|---|---|---|---|---|---|---|
| p00 | 5,727.88 | 2,277.72 | 3,450.17 | 1,816.74 | 0.023033 | 0.005916 | 4.48e-08 | True |
| p25 | 6,104.25 | 2,195.81 | 3,908.43 | 1,860.05 | 0.022640 | 0.006526 | 1.14e-05 | True |
| p50 | 5,979.23 | 2,103.78 | 3,875.45 | 1,904.46 | 0.023645 | 0.005888 | 1.82e-07 | True |
| p75 | 6,142.67 | 1,904.88 | 4,237.79 | 2,032.60 | 0.025527 | 0.005086 | 2e-06 | True |

All four implemented signed-rank calculations were independently cross-checked against SciPy 1.17.1 using the same two-sided asymptotic, tie-corrected convention. SciPy was used for verification, not as a hidden optimizer dependency. The condition-wise tests follow the stated protocol; no additional family-wide claim across all conditions or descriptive k values is asserted.

The p = 0 improvement over equal weights is especially important for interpretation. Purpose still determines base shares in that control, while household adjustment is disabled. Learned weights can improve by emphasizing purpose and suppressing irrelevant variation. Therefore this comparison alone cannot show that the search recovered the household relationship. The purpose baseline remains the relevant no-household comparison; the primary pooled learned peso error remains above it. Any small subgroup fluctuation below that baseline is not, on its own, proof of leakage or of a real household effect.

### Gain across household-signal levels

The thirty-set scaled gains are .023033, .022640, .023645 and .025527 at p = 0, .25, .50 and .75. Their spread is **0.002887**, smaller than the mean replicate SD of **0.005854**. The gain is descriptively flat rather than showing a clear increase with household signal. This range-to-SD comparison is a descriptive diagnostic, not a formal trend or equivalence test. Together with improvement at p = 0 and the weight-mass results below, it identifies purpose recovery without detectable household-pattern recovery at any tested signal level under this implementation and evidence. It does not prove that no household information could be recovered by any method.

## 4.8 Weight recovery and reference gap

The written household score uses 28 matching fields. Purpose also determines the base share, so the amount-rule reference includes 29 matching fields. It assigns equal positive weight to those 29 and zero to the other matching fields. Cost is not a matching field and cannot enter this reference. This is a fixed nominal rule-field diagnostic across conditions; p = 0 disables the household term. Both purpose-inclusive and household-only weight masses are shown to avoid mistaking purpose recovery for household recovery. The reference is not a proven optimum. A negative learned-minus-reference gap is possible and is not an error.

| Condition | Mean rule-field weight % | Mean household-only weight % | Mean zero-rule fields in top 10 | Learned scaled error | Reference scaled error | Learned minus reference |
|---|---|---|---|---|---|---|
| p00 | 79.45 | 71.45 | 2.20 | 0.156962 | 0.176583 | -0.019620 |
| p25 | 76.87 | 67.98 | 2.60 | 0.145766 | 0.167306 | -0.021540 |
| p50 | 79.25 | 71.61 | 1.80 | 0.138103 | 0.156741 | -0.018638 |
| p75 | 77.28 | 69.12 | 2.00 | 0.125454 | 0.145463 | -0.020009 |

## Household weight mass did not increase

| p = .50 household-only diagnostic | Weight mass |
|---|---|
| Equal weights on the 28 household-score fields | **71.79%** |
| Learned weights on the same fields | **71.61%** |
| Change | **−0.18 percentage points** |

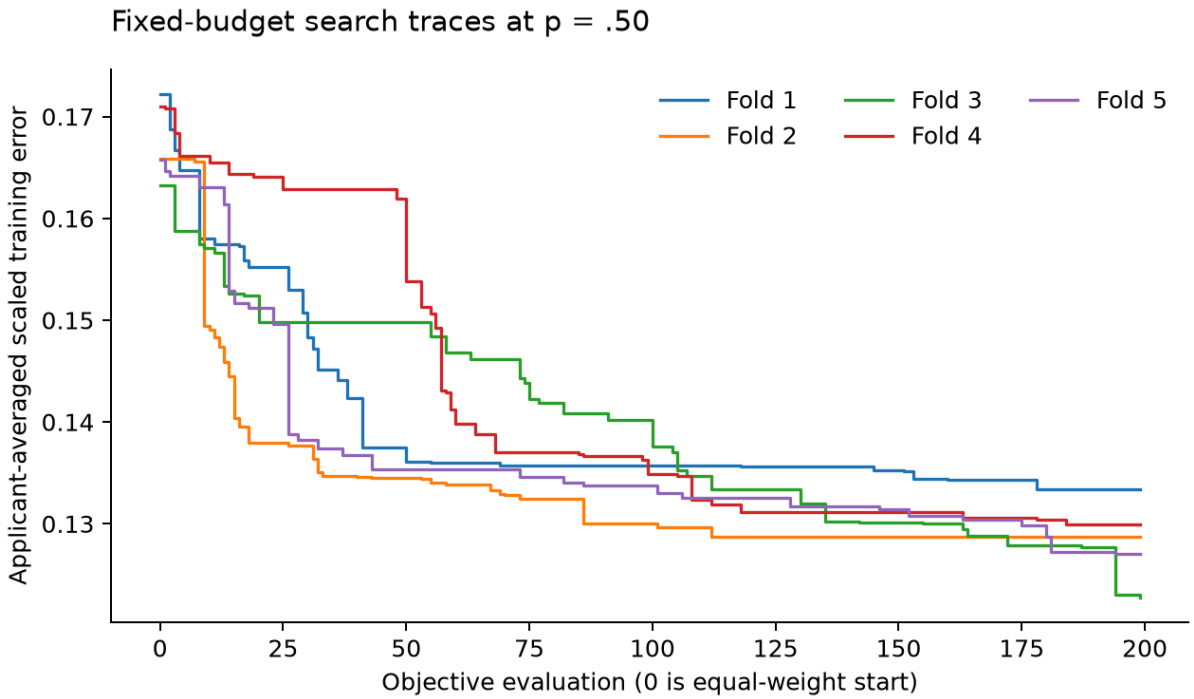
Equal weights allocate 74.36% to all 29 nominal rule fields including purpose. The larger learned rule-field mass is driven chiefly by purpose, not increased household mass. Household-only mass is also below the equal-weight share at the other three signal levels in the table. **The weight search recovers purpose and does not detectably recover the household pattern at any tested signal level.** This is a finding about the present search and diagnostics; aggregate weight mass alone cannot rule out every redistribution among household fields.

Largest mean weights at p = .50 (all 39 weights and every candidate trace are retained in the evidence archive):

| Field | Mean weight | In nominal amount rule |
|---|---|---|
| purpose_of_assistance | 2.9797 | True |
| employed_members | 1.5750 | True |
| ext_mss_discount | 1.4516 | True |
| ext_senior_citizen_discount | 1.3562 | True |
| total_income_past_6_months | 1.3125 | True |
| vuln_unstable_employment | 1.2688 | True |
| support_employer | 1.2625 | True |
| seasonal_members | 1.2000 | False |
| crisis_disablement | 1.1812 | True |
| ext_philhealth | 1.1500 | True |

Each search stopped at its fixed budget, not a claim of convergence to an optimum. The trace reports the incumbent after each candidate evaluation.

| Fold | Initial scaled objective | Final scaled objective | Last accepted evaluation | Accepted moves |
|---|---|---|---|---|
| 1 | 0.172210 | 0.133434 | 178 | 25 |
| 2 | 0.165831 | 0.128744 | 112 | 26 |
| 3 | 0.163281 | 0.122800 | 199 | 32 |
| 4 | 0.171023 | 0.129934 | 184 | 30 |
| 5 | 0.165737 | 0.127070 | 194 | 26 |




## 4.9 Sensitivity to the number of retrieved cases

The following applicant-averaged scaled errors vary k while keeping the selected settings and learned weights fixed. They are descriptive, with no retuning and no additional superiority claim. The main test and fitting objective remain k=5.

| Condition | k | Equal scaled MAE | Learned scaled MAE |
|---|---|---|---|
| p00 | 3 | 0.189847 | 0.164333 |
| p00 | 5 | 0.182232 | 0.156962 |
| p00 | 7 | 0.179067 | 0.155340 |
| p00 | 10 | 0.173872 | 0.149234 |
| p25 | 3 | 0.183085 | 0.155071 |
| p25 | 5 | 0.171892 | 0.145766 |
| p25 | 7 | 0.164606 | 0.138290 |
| p25 | 10 | 0.160377 | 0.133942 |
| p50 | 3 | 0.174302 | 0.144091 |
| p50 | 5 | 0.167927 | 0.138103 |
| p50 | 7 | 0.159000 | 0.137273 |
| p50 | 10 | 0.157264 | 0.133901 |
| p75 | 3 | 0.162276 | 0.131079 |
| p75 | 5 | 0.153894 | 0.125454 |
| p75 | 7 | 0.150651 | 0.121224 |
| p75 | 10 | 0.148152 | 0.119983 |

## 4.10 Computational performance and reproducibility

Retrieval timing used three warmups and eleven timed calls for one fixed reviewed query per assistance type, starting from the complete 1,000-event reference set. These are function timings, not end-to-end network latency or a population-wide latency estimate. Memory is the process working set after

importing the backend and loading the default CSV, measured in five fresh processes, not peak memory.

| Type | Query | Median retrieval ms |
|---|---|---|
| Cash Relief Assistance | J07 | 2.595 |
| Educational | J02 | 2.127 |
| Food Assistance | J06 | 2.120 |
| Funeral | J04 | 3.771 |
| Medical | J01 | 11.503 |
| Transportation | J08 | 2.219 |

Median backend working set was 49.24 MiB. The full five-fold, four-condition weight evaluation had median runtime 38.29 seconds across three deterministic repetitions (36.11, 38.56, 38.29 seconds). The repeated learned outcomes and weights were identical. This time includes distance construction, search, held-out evaluation and trace writes; it excludes the thirty replicate runs, input generation, final document drafting, and the later reporting-only reference/count supplement.

The recorded machine used 12th Gen Intel(R) Core(TM) i7-12700H, 15.63 GiB reported physical memory, Microsoft Windows 11 Home Single Language version 10.0.26200, and Python 3.12.10. The tested frontend used React 19.3.0 and Vite 7.3.6. Exact installed Python versions and the frontend lock file accompany the run. The configured repository is `https://github.com/HenriDillo/aics-case-based-retrieval.git`; its private status was authenticated and verified on 19 October 2026. No external push or message was performed by this work.

## 4.11 Prototype and review status

The prototype loads the reconciled p = .50 synthetic reference CSV once, retrieves five same-type cases using equal weights, and displays their circumstances, documented costs and generated grants. Browser checks across all six assistance types covered retrieval, numeric grant display, comparison navigation and mobile overflow. The production frontend build passed. Software verification establishes implementation behavior; it does not change the failed research success criterion.

The printable CSV-driven packet fixes funds at ₱250,000 and includes populated diagnoses for every assistance type. All forty supplied reviewed cases and all their nonempty displayed values were checked. The packet has two printable pages per case; the original reviewed DOCX remains unchanged and available separately. No private pair key is included in the reviewer packet.

No new DSWD response to this completed run or packet has been supplied or verified. The earlier consultations described in Chapters 1–3 are not represented as approval of these new results. The record therefore cannot claim that the new packet was submitted, accepted, rejected, or ignored. The four review topics remain field ranges, assistance-type mix, amount spread and implausible combinations. Actual feedback can be added with its date and source when available.

## 4.12 Interpretation and limitations

The fixed reviewed-case test does not support the proposed retrieval method's success on this generated data under its preregistered peso-error criterion. Separately, learned weights improve over equal weights within the implemented synthetic experiments. The control result and the large role of purpose

limit an interpretation based on household-pattern recovery. Statistical improvement in scaled error is also distinct from passing the pooled peso test.

The data-generating mechanism includes assumed costs, a flattened type mix, chosen missingness, historical income thresholds and a known amount rule. Its source scoring treats some unknown inputs as absence or zero, while retrieval does not impute them. These choices restrict external validity. The forty-case test has small type-specific counts, so those rows should not be treated as precise type-level estimates. The fixed greedy search budget is reproducible but does not guarantee globally best weights. No result here estimates the accuracy, fairness or operational usefulness of decisions on actual DSWD applicants.

The complete evidence retains all seeds, fold assignments, event predictions, applicant errors, comparison counts, candidate traces and source hashes. Failed results and unverified external feedback remain visible. The chapter is ready for the research team's substantive review, not presented as an approved thesis chapter or evidence of successful real-world deployment.

### Unknown-value defects and the decision required before a rerun

The amount rule treats some unknowns as absence or zero, whereas Gower omits fields missing in either record. Unknown emergency fund, savings and insurance values can therefore attract absence points, and an unreadable employed-member count can be treated as zero workers. Unknown six-month income returns zero band points, the same numeric contribution as the highest income band: this affects 52 primary-base events. This mismatch can encode amount variation that retrieval cannot compare as an observed household attribute.

The proposed correction is to omit unknown contributions with an explicitly agreed available-field aggregation rule. Simply replacing the missing-income return with zero or skipping an addition to an unnormalised sum would leave that income contribution numerically unchanged; the aggregation and recalibration must be specified before approval. No corrected amounts or new comparisons are included here.

The 19 October brief requires Ma’am Grace’s dated written agreement before correcting the data and rerunning the original same-type test, or running the proposed same-purpose test. The proposed change would revise rules eight and eleven in Section 3.2.2 so numeric ranges use the revised eligible pool. The ≥20% criterion, Section 3.2.4.2 evidence standard, seeds, k, search budget, weights procedure, bands and noise remain fixed. Both corrected-data tests must be reported, alongside the retained prior runs, whether they pass or fail. Same-purpose pools also have thin medical subgroups: assistive device n = 14, chemotherapy n = 18 and dialysis n = 19 before exclusions. Approval has not been supplied and neither comparison has been run.

## Appendix 4A Per-type case-base errors

Each row first averages event errors within applicant and assistance type, then averages those applicant errors. An applicant with events in more than one type can appear in more than one row. Peso and scaled errors are both reported; transportation has no defined scaled width. The purpose baseline uses the same eligible pool and exclusions as retrieval.

| Condition | Method | Type | Applicants | Events | MAE ₱ | Baseline ₱ | Scaled MAE | Reduction % |
|---|---|---|---|---|---|---|---|---|
| p00 | equal | Cash Relief Assistance | 98 | 100 | 1,106.63 | 968.37 | 0.122959 | -14.28 |
| p00 | equal | Educational | 78 | 80 | 1,497.48 | 1,449.27 | 0.332773 | -3.33 |
| p00 | equal | Food Assistance | 80 | 80 | 851.25 | 750.00 | 0.425625 | -13.50 |
| p00 | equal | Funeral | 143 | 150 | 2,925.54 | 2,726.14 | 0.146277 | -7.31 |
| p00 | equal | Medical | 461 | 510 | 9,029.69 | 1,961.14 | 0.141213 | -360.43 |
| p00 | equal | Transportation | 80 | 80 | 0.00 | 0.00 | N/A | N/A |
| p00 | learned | Cash Relief Assistance | 98 | 100 | 1,087.24 | 968.37 | 0.120805 | -12.28 |
| p00 | learned | Educational | 78 | 80 | 1,514.63 | 1,449.27 | 0.336585 | -4.51 |
| p00 | learned | Food Assistance | 80 | 80 | 930.62 | 750.00 | 0.465313 | -24.08 |
| p00 | learned | Funeral | 143 | 150 | 3,048.22 | 2,726.14 | 0.152411 | -11.81 |
| p00 | learned | Medical | 461 | 510 | 2,649.67 | 1,961.14 | 0.088503 | -35.11 |
| p00 | learned | Transportation | 80 | 80 | 0.00 | 0.00 | N/A | N/A |
| p25 | equal | Cash Relief Assistance | 98 | 100 | 981.89 | 889.29 | 0.109099 | -10.41 |
| p25 | equal | Educational | 78 | 80 | 1,568.83 | 1,540.58 | 0.348629 | -1.83 |
| p25 | equal | Food Assistance | 80 | 80 | 714.38 | 632.50 | 0.357187 | -12.94 |
| p25 | equal | Funeral | 143 | 150 | 3,050.22 | 2,900.77 | 0.152511 | -5.15 |
| p25 | equal | Medical | 461 | 510 | 9,576.65 | 1,964.81 | 0.135346 | -387.41 |
| p25 | equal | Transportation | 80 | 80 | 0.00 | 0.00 | N/A | N/A |
| p25 | learned | Cash Relief Assistance | 98 | 100 | 1,023.47 | 889.29 | 0.113719 | -15.09 |
| p25 | learned | Educational | 78 | 80 | 1,795.99 | 1,540.58 | 0.399109 | -16.58 |
| p25 | learned | Food Assistance | 80 | 80 | 656.25 | 632.50 | 0.328125 | -3.75 |
| p25 | learned | Funeral | 143 | 150 | 3,029.14 | 2,900.77 | 0.151457 | -4.43 |
| p25 | learned | Medical | 461 | 510 | 2,496.94 | 1,964.81 | 0.083826 | -27.08 |
| p25 | learned | Transportation | 80 | 80 | 0.00 | 0.00 | N/A | N/A |
| p50 | equal | Cash Relief Assistance | 98 | 100 | 921.43 | 927.55 | 0.102381 | 0.66 |
| p50 | equal | Educational | 78 | 80 | 1,383.55 | 1,535.44 | 0.307455 | 9.89 |
| p50 | equal | Food Assistance | 80 | 80 | 847.50 | 836.25 | 0.423750 | -1.35 |
| p50 | equal | Funeral | 143 | 150 | 2,451.38 | 2,539.45 | 0.122569 | 3.47 |
| p50 | equal | Medical | 461 | 510 | 9,609.42 | 2,181.87 | 0.130962 | -340.42 |
| p50 | equal | Transportation | 80 | 80 | 0.00 | 0.00 | N/A | N/A |
| p50 | learned | Cash Relief Assistance | 98 | 100 | 906.12 | 927.55 | 0.100680 | 2.31 |
| p50 | learned | Educational | 78 | 80 | 1,458.11 | 1,535.44 | 0.324025 | 5.04 |
| p50 | learned | Food Assistance | 80 | 80 | 815.00 | 836.25 | 0.407500 | 2.54 |
| p50 | learned | Funeral | 143 | 150 | 2,207.77 | 2,539.45 | 0.110388 | 13.06 |
| p50 | learned | Medical | 461 | 510 | 2,626.08 | 2,181.87 | 0.080893 | -20.36 |
| p50 | learned | Transportation | 80 | 80 | 0.00 | 0.00 | N/A | N/A |
| p75 | equal | Cash Relief Assistance | 98 | 100 | 833.93 | 926.79 | 0.092659 | 10.02 |
| p75 | equal | Educational | 78 | 80 | 1,241.37 | 1,488.32 | 0.275859 | 16.59 |
| p75 | equal | Food Assistance | 80 | 80 | 704.38 | 915.00 | 0.352187 | 23.02 |
| p75 | equal | Funeral | 143 | 150 | 2,411.05 | 2,830.06 | 0.120552 | 14.81 |
| p75 | equal | Medical | 461 | 510 | 9,991.49 | 2,309.18 | 0.125976 | -332.69 |
| p75 | equal | Transportation | 80 | 80 | 0.00 | 0.00 | N/A | N/A |
| p75 | learned | Cash Relief Assistance | 98 | 100 | 803.06 | 926.79 | 0.089229 | 13.35 |
| p75 | learned | Educational | 78 | 80 | 1,305.22 | 1,488.32 | 0.290048 | 12.30 |
| p75 | learned | Food Assistance | 80 | 80 | 763.75 | 915.00 | 0.381875 | 16.53 |
| p75 | learned | Funeral | 143 | 150 | 2,421.29 | 2,830.06 | 0.121065 | 14.44 |
| p75 | learned | Medical | 461 | 510 | 2,258.18 | 2,309.18 | 0.067602 | 2.21 |
| p75 | learned | Transportation | 80 | 80 | 0.00 | 0.00 | N/A | N/A |

## Sources and evidence

- Manuscript Chapters 1–3 v25, supplied by the research team, especially Sections 1.5 and 3.2.1–3.2.4 and Table 3.1.
- Context for Astra, 18 October 2026, and Amount Rule Specification v3, supplied in the AICS v25 handoff.
- DSWD PDPB (2022), Process Evaluation of AICS, historical cost/share benchmarks transcribed in the supplied code and checked source tables. Device and transport values are explicitly labelled assumptions, not additional published estimates.
- Reconciliation contract, method choices, generated-data manifests and the completed-study-v1 machine-readable results accompanying this draft.